



# 集成电路技术工艺基础

## Final review

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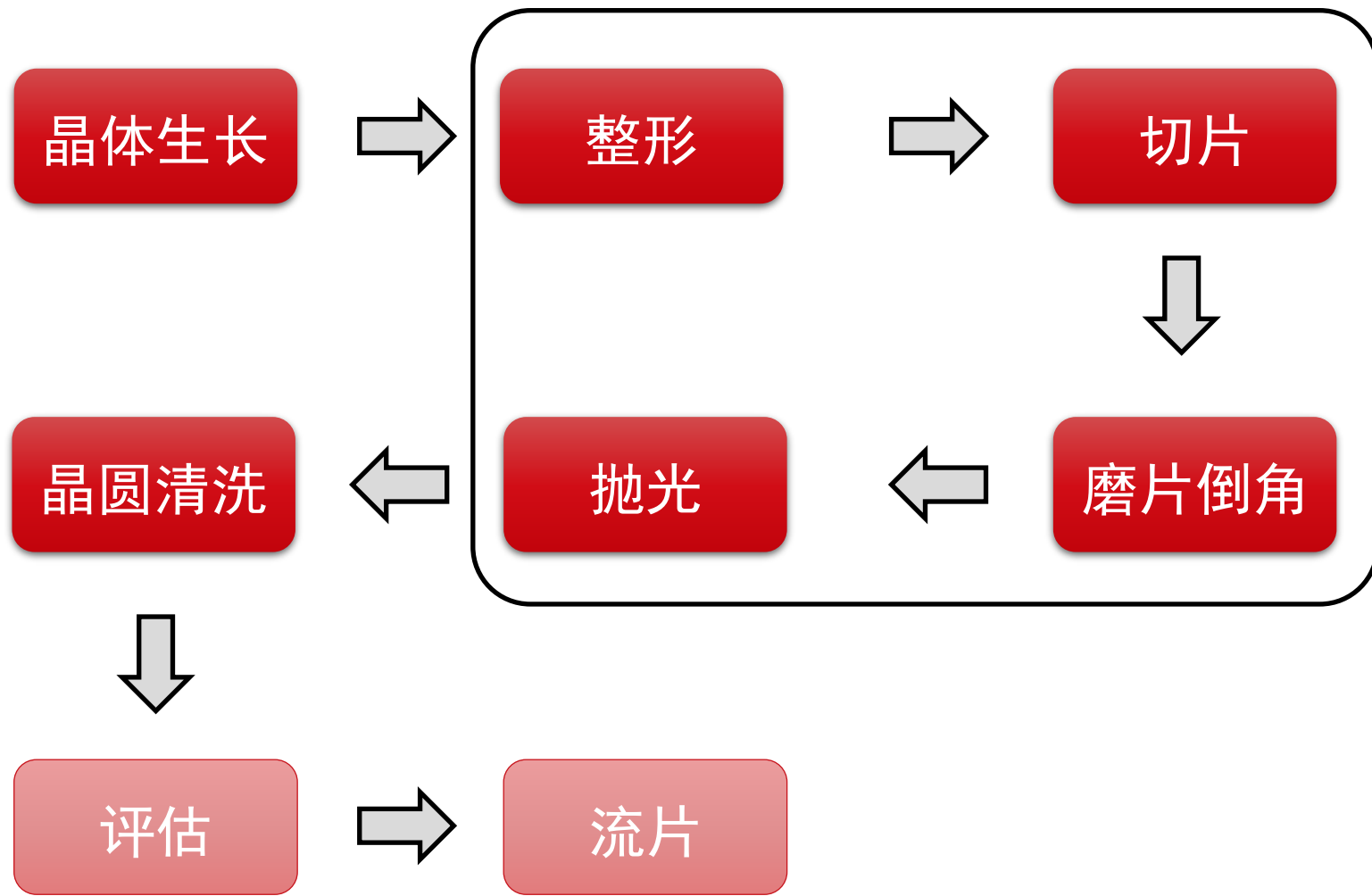


- ✎ **Chapter 2 Si crystal growth and wafer manufacturing**
- ✎ **Chapter 3 Oxidation of Silicon**
- ✎ **Chapter 4 Diffusion**
- ✎ **Chapter 5 Implantation**
- ✎ **Chapter 10 CMOS process**

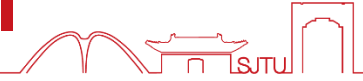
# 硅晶圆生长、制造流程



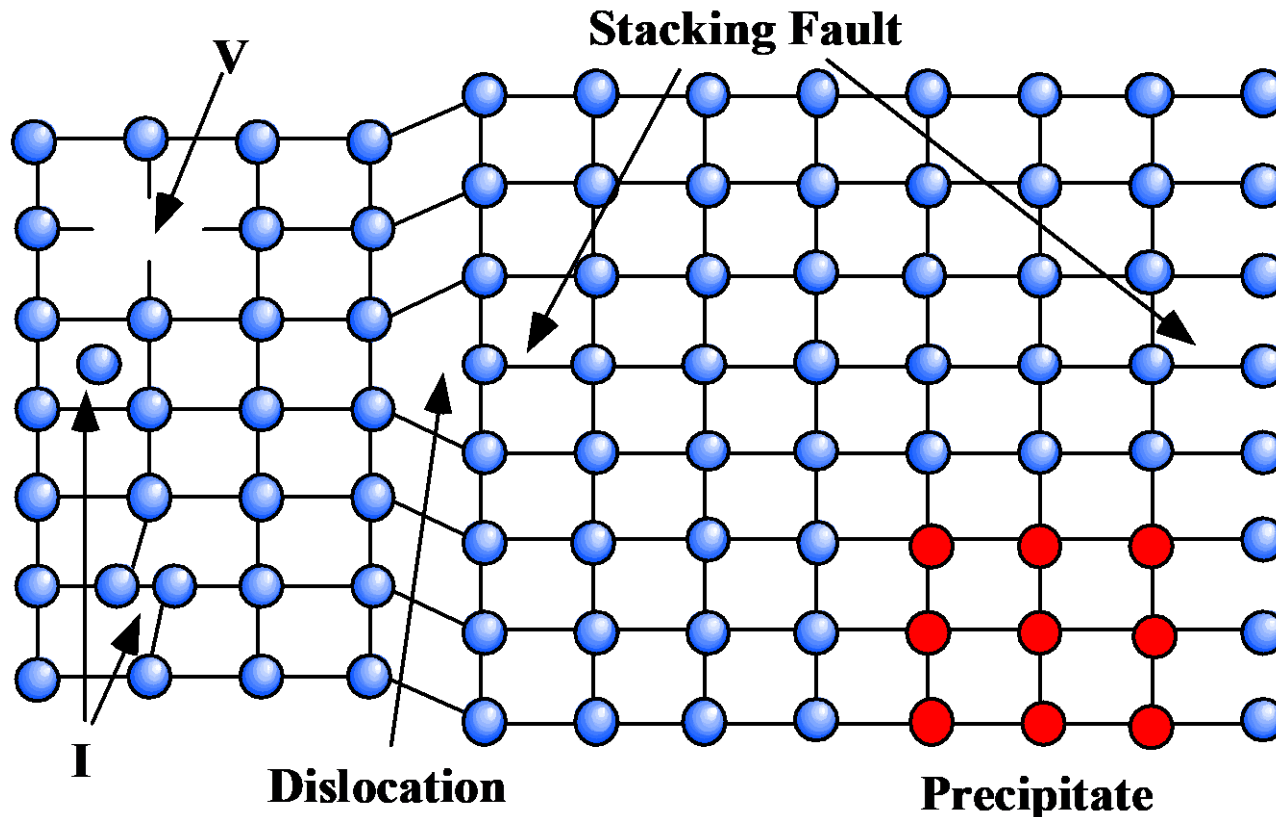
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# 2.1 Structure and defects in Si crystal



- Point defects: vacancy, interstitial, substitutional, Frankel defects
- Linear defects: edge and screw dislocations
- Planar defects: stacking fault, grain boundaries, surfaces
- Bulk defects: cracks and pores, precipitate



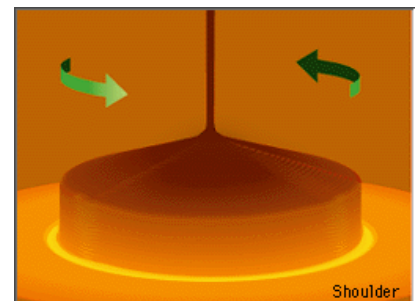
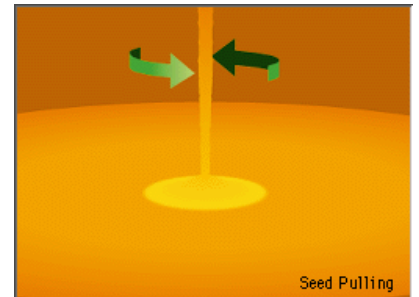
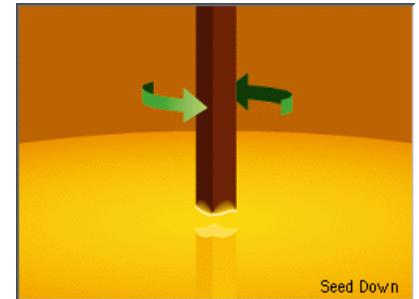
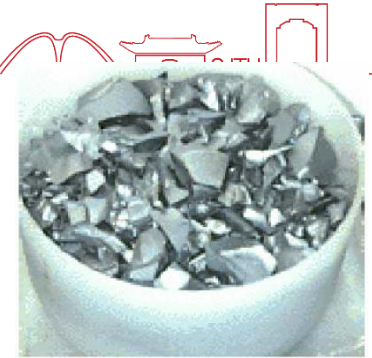
**These defects are generally detrimental to device performance and also impact the behavior of processes (e.g. dopant diffusion)**

## 2.2 Single crystal growth

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- EGS is broken into small pieces and placed in an  $\text{SiO}_2$  crucible. In an Ar ambient, the crucible is heated to just above  $1417^\circ\text{C}$ .
- Dopant is added to the melt to intentionally dope the resulting crystal
- A single crystal seed is then lowered into the melt and withdrawn slowly. Melt flows up the seed and cools as crystal begins to grow.
- Seed rotated about its axis to produce a circular cross-section crystal. The rotation inhibits the natural tendency of the crystal to grow along certain orientations to produce a faceted crystal.
- Long ingots (錠) (boules)  $\sim 100$  kg, with very good circular cross-section are produced.

**Pull rate, melt temperature and rotation rate are all important control parameters.**



## 2.3 Main process of wafer manufacturing



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1. EGM  
fabrication



2. Single crystal  
growth



3. diameter  
grinding



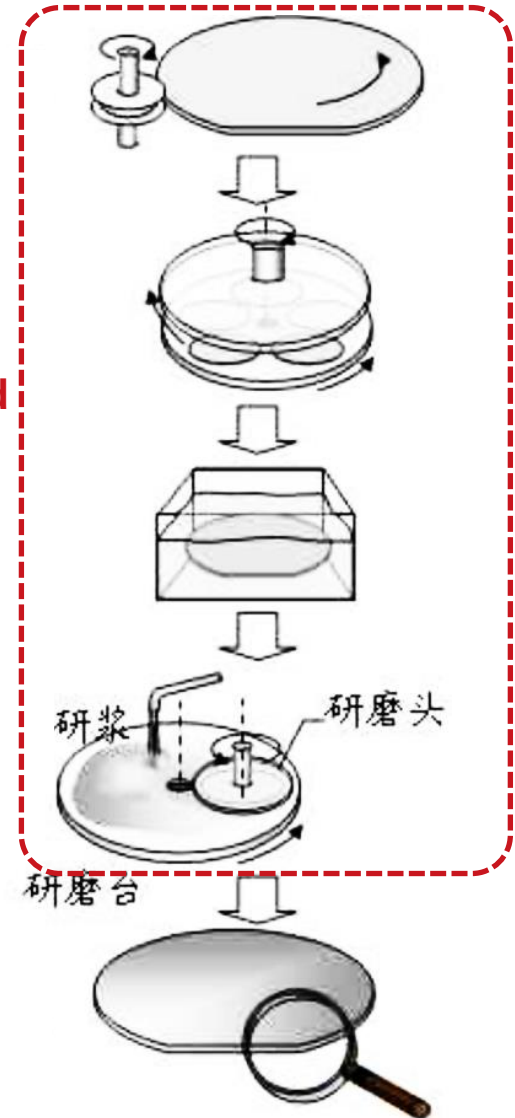
4. Main plane  
formation



5. Slicing



6. Wafer  
polishing and  
marking



7. Wafer  
evaluation

### Cleaning room + Cleaning process

#### Organic/resistor contamination Reduction

##### ➤ Dry cleaning

Oxygen plasma

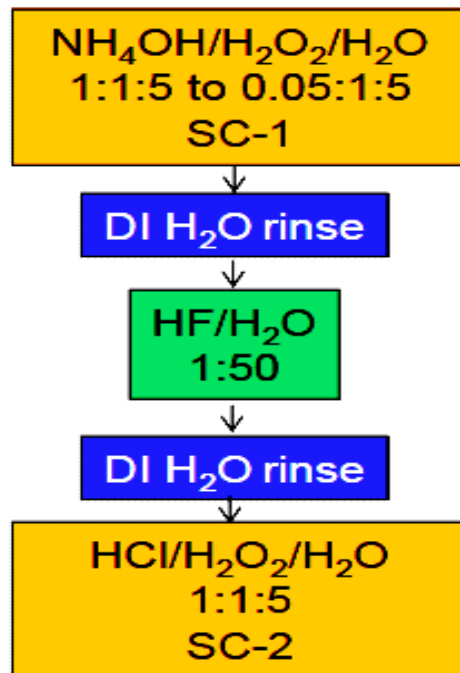
Decompose resistor to  $\text{CO}_2 + \text{H}_2\text{O}$

##### ➤ Wet cleaning

$\text{H}_2\text{SO}_4(98\%):\text{H}_2\text{O}_2(30\%) = 2:1-4:1$

Decompose organic to  $\text{CO}_2 + \text{H}_2\text{O}$

#### RCA process to remove organics, heavy metals and alkali ions



##### ➤ SC-1 (APM: Ammonia Peroxide Mixture)

-70-80° C/10min

-Remove organics, and particles

-Etches Si surface very slowly

##### ➤ Remove oxide

-Room T/1min

##### ➤ SC-2 (HPM: Hydrochloric Peroxide Mixture)

-70-80° C/10min

-Remove alkali ions such as  $\text{Al}^{3+}$ ,  $\text{Fe}^{3+}$ ,  $\text{Mg}^{2+}$

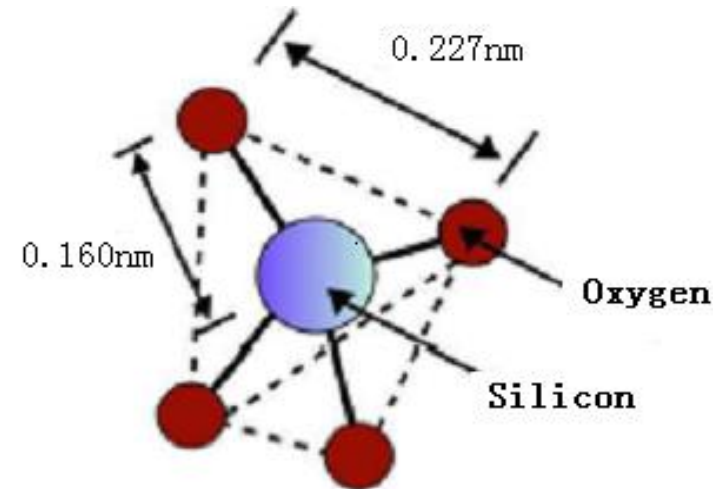
-Remove heavy metals like Au

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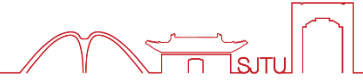
# 3. 1 Why is oxidation of Si needed?

- $\text{SiO}_2$  is stable down to  $10^{-9}$  Torr,  $T > 900^\circ\text{C}$
- $\text{SiO}_2$  is a diffusion barrier for B, P, As
- $\text{SiO}_2$  can be etched with HF which leaves Si unaffected
- $\text{SiO}_2$  is good insulator,  $E_g > 8\text{eV}$
- $\text{SiO}_2$  has high dielectric breakdown field,  $500\text{V}/\mu\text{m}$
- $\text{SiO}_2$  grown on Si  $\rightarrow$  clean Si/ $\text{SiO}_2$

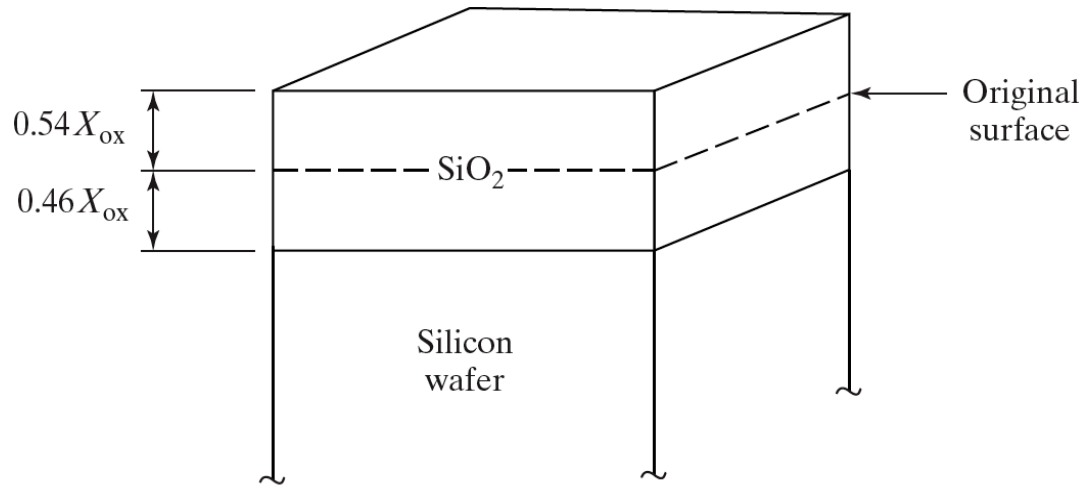


**$\text{SiO}_2$ 是上帝赐给集成电路的材料 (天选之材)**

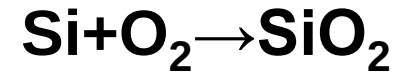
## 3.2 Method of thermal oxidation



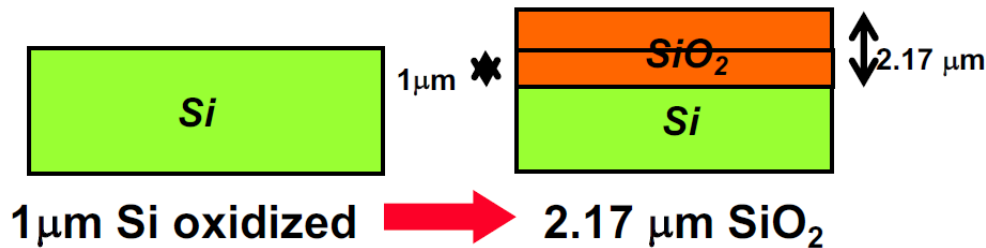
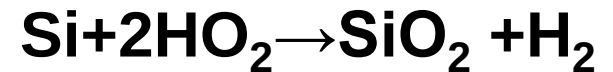
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**Dry oxidation**



**Wet oxidation**



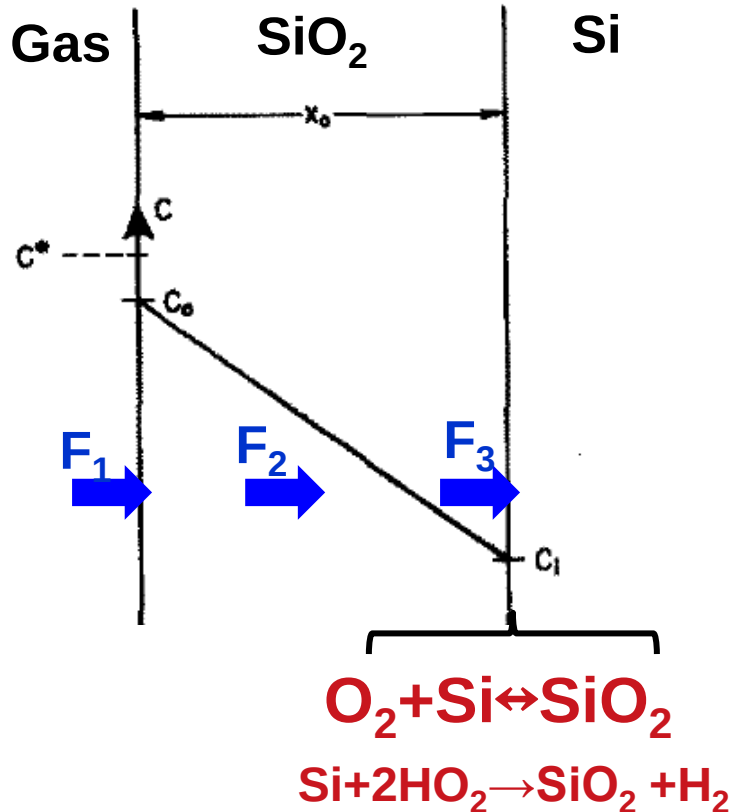
$$X_{\text{si}} = X_{\text{ox}} \cdot \frac{N_{\text{ox}}}{N_{\text{si}}}$$

**Growth Occurs 54% above and 46% below original surface as silicon is consumed.**

### 3.3 Kinetics of oxidation



**F: flux** – the number of oxygen molecules that crosses a plane per unit area per second



**At surface**

$$F_1 = h(C^* - C_o) \quad h_g \text{ 是质量输运系数}$$

**h: Mass transfer coefficient [cm/sec].**

**In SiO<sub>2</sub>    D<sub>eff</sub>: Diffusivity [cm<sup>2</sup>/sec]**

$$F_2 = D_{eff} \frac{(C_o - C_i)}{x} \quad \text{菲克第一定律}$$

**At SiO<sub>2</sub>/Si interface**

$$F_3 = kC_i \quad \text{k: Oxidation reaction rate constant}$$

**Oxidant diffusion → interface reaction.**

### 3.3 Deal-Grove Model



$$x = \frac{A}{2} \left( \sqrt{1 + \frac{t + \tau}{A^2/4B}} - 1 \right)$$

When  $t \ll A^2/4B$ 时,

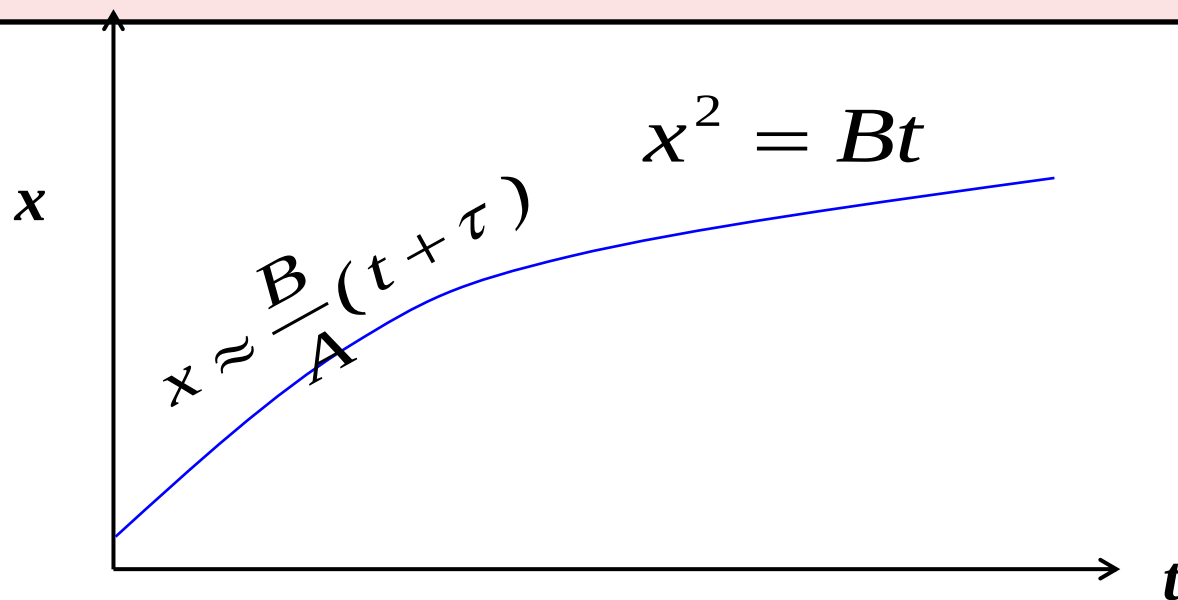
$$x \approx \frac{B}{A}(t + \tau)$$

(线性规律, 反应控制)

When  $t \gg A^2/4B$ ,

$$x^2 \approx B(t + \tau)$$

(抛物型规律, 扩散控制)



$$\sqrt{1+x}-1 \sim \frac{x}{2}$$

- **Partial pressure of oxidant (氧化剂分压)**
- **Temperature of oxidation (氧化温度)**
- **Surface orientation of Si (硅晶面取向)**
- **Doping and Impurity (杂质)**
- **Dry and Wet oxidation (干湿法氧化)**

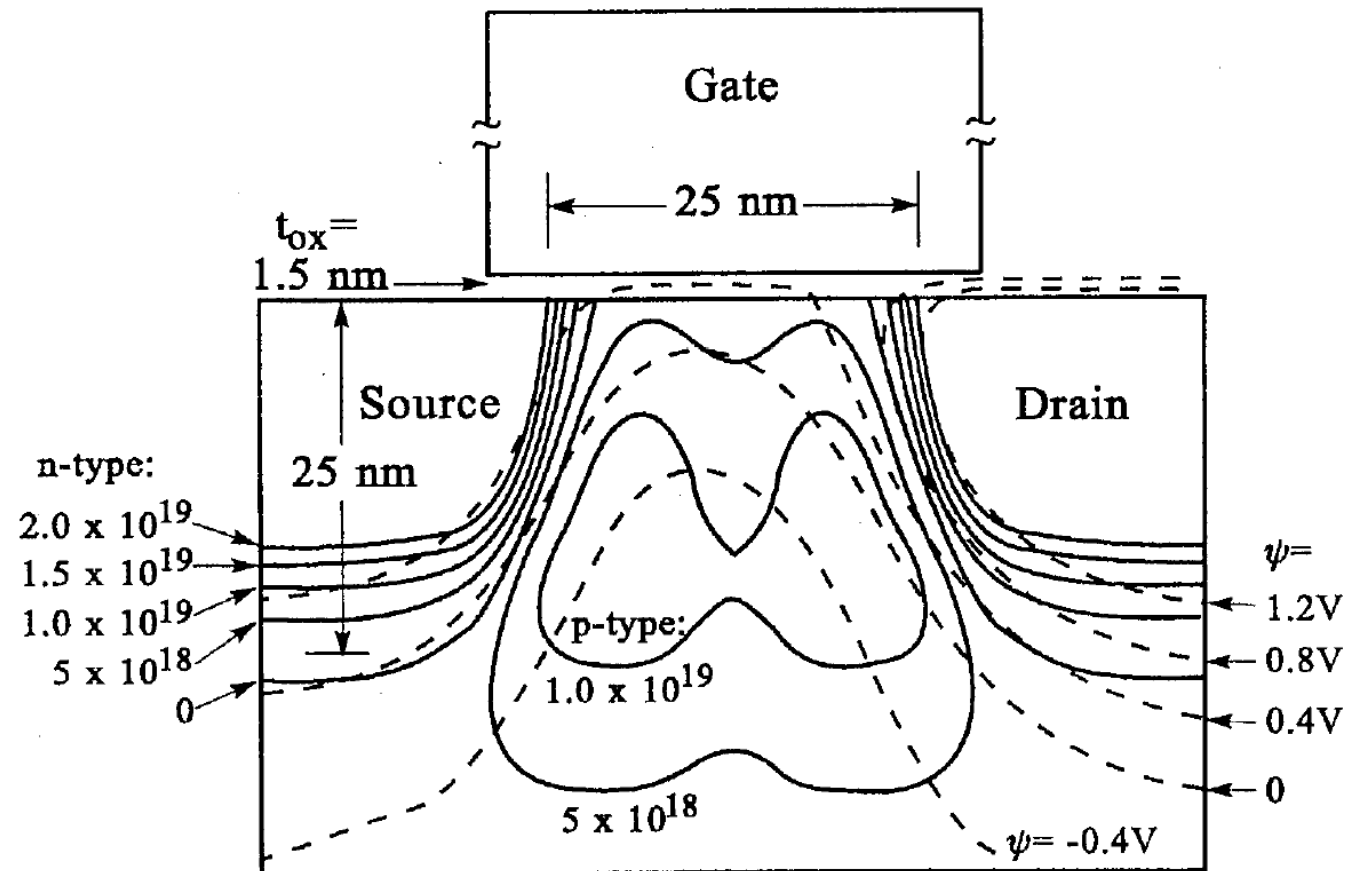
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# 4.1. Why is diffusion needed?



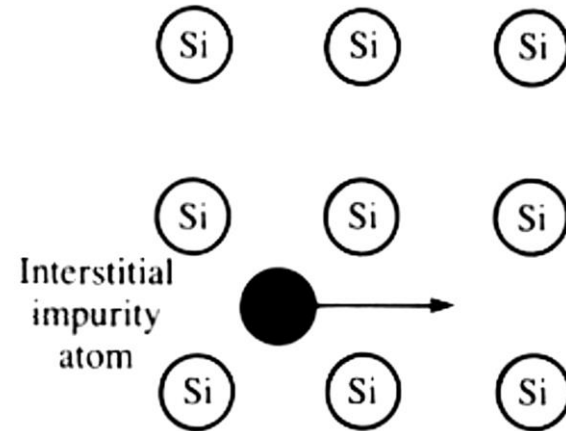
## Halo or Pocket Profiles: Improve Manufacturability

25 nm Gate Length MOSFET Design



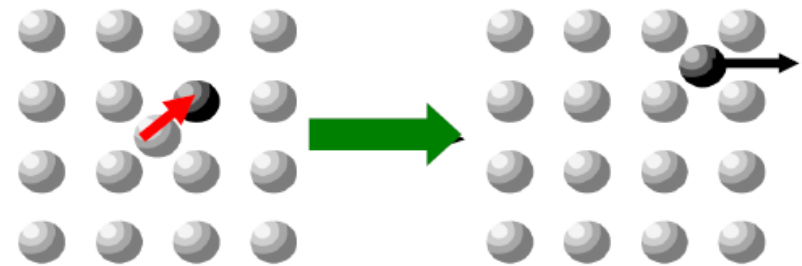
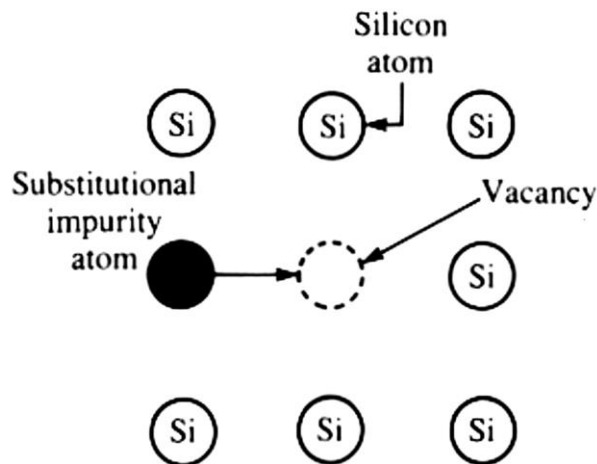
## (1) Interstitial Diffusion

Example: Cu, Fe, Li, H



## (b) Substitutional Diffusion      (c) Interstitialcy Diffusion

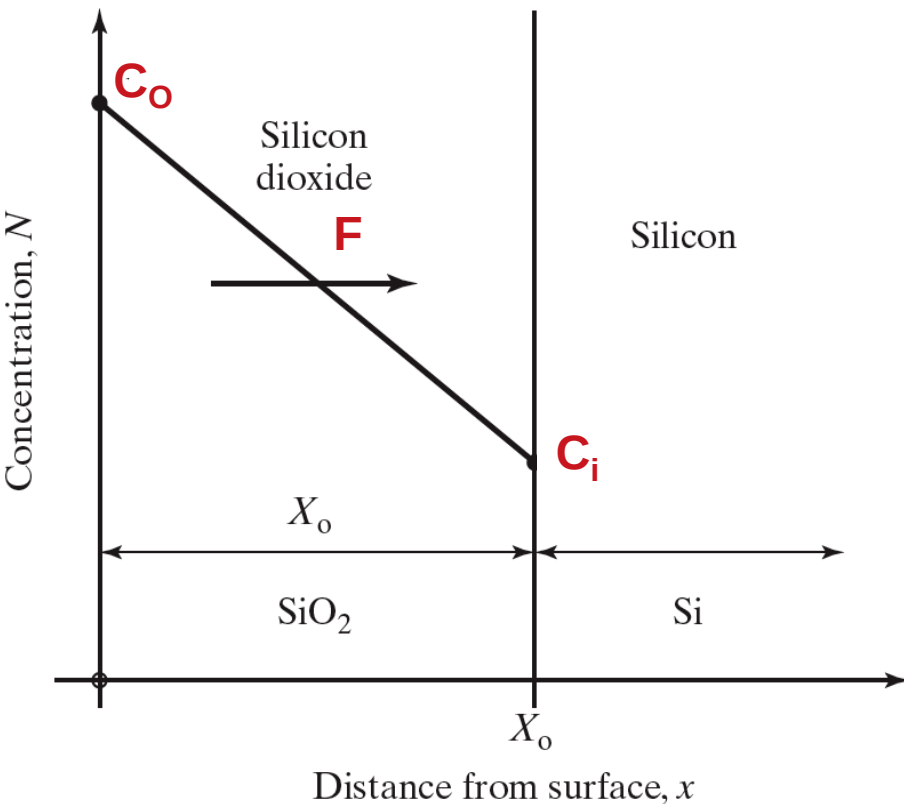
Example: Dopants in Si ( e.g. B, P,As,Sb)





### Fick's first law:

How the flux of dopant depends upon the doping gradient:



$$F = -\frac{DdC}{dx}$$

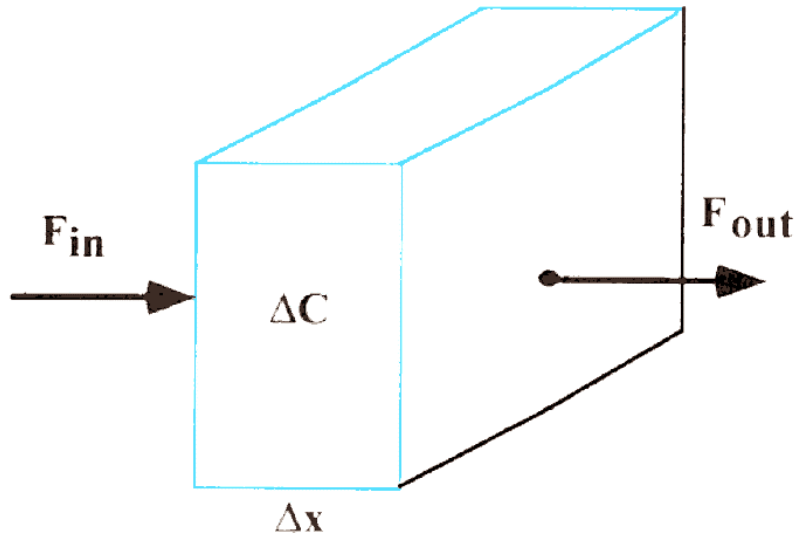
**D = diffusion coefficient**

**Flux**– the number of atoms that crosses a plane per unit area per second

When concentration gradient goes to zero, the flow stops.

### Fick's Second Law

how the change in concentration in a volume element is determined by the change in fluxes in and out of the volume.



$$\frac{\Delta C}{\Delta t} \Delta x = -(F_{out} - F_{in})$$

$$\frac{\partial C}{\partial t} = -\frac{\partial F}{\partial x} = \frac{\partial}{\partial x} \left( D \frac{\partial C}{\partial x} \right)$$

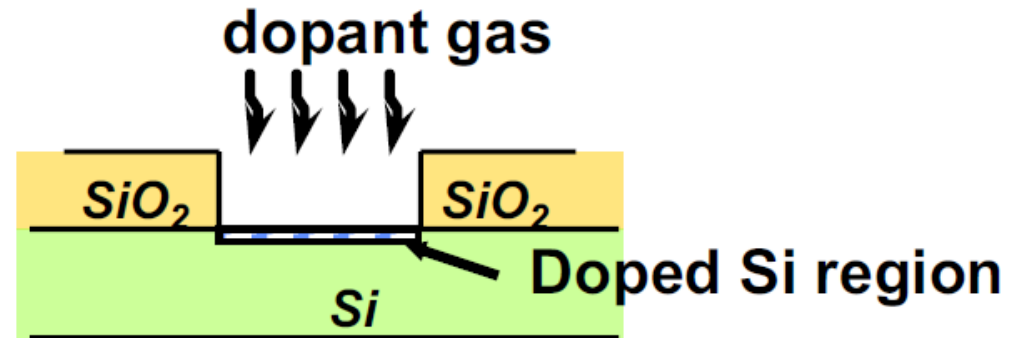
$$\frac{dC}{dt} = \frac{D d^2 C}{dx^2}$$

## 4.3 Two step Dopant Diffusion

IC fabrication processes involve a number of different steps

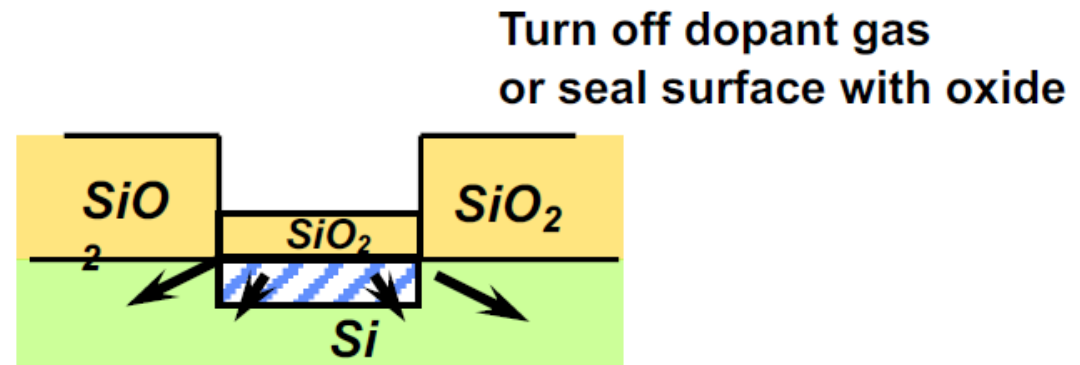
### (1) Predeposition

dose control



### (2) Drive-in

profile control  
(junction depth;  
concentration)



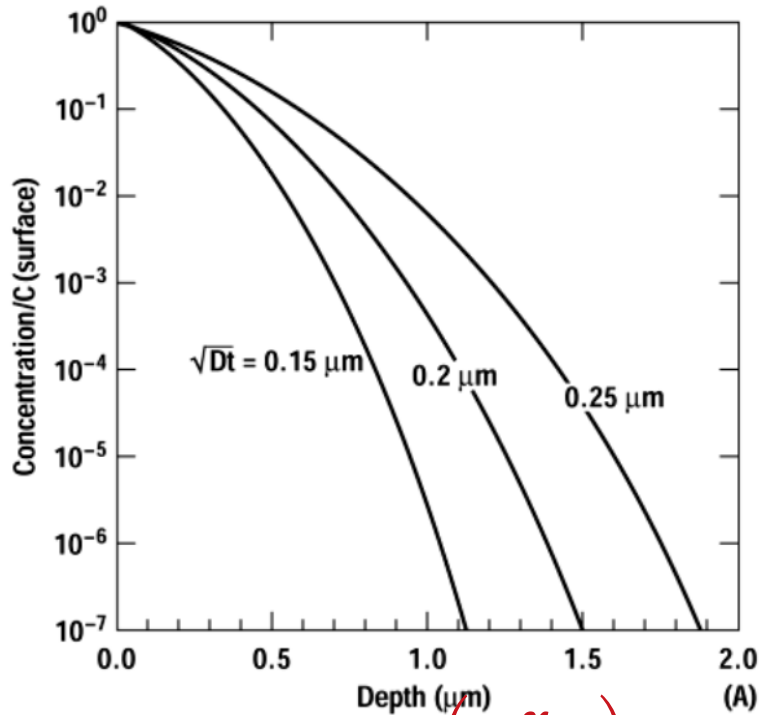
**Note: Predeposition by diffusion can be replaced by a shallow implantation step.**

# 4.3 Two step Dopant Diffusion



Constant surface concentration

**Predeposition**

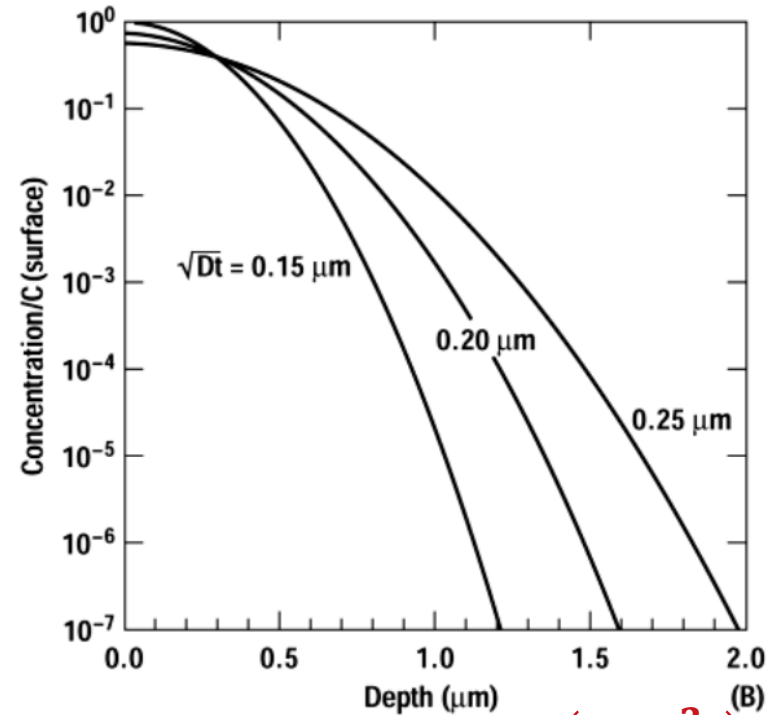


$$C(x, t) = C_0 \operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$Q = \int_0^{\infty} C(x, t) dx = 2 C_0 \sqrt{\frac{Dt}{\pi}}$$

Constant dose

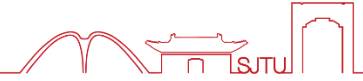
**Drive-in**



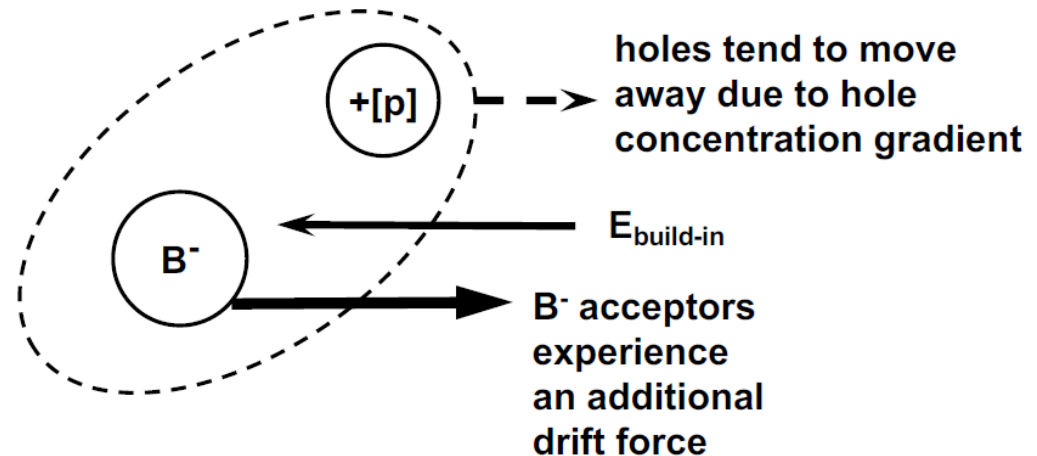
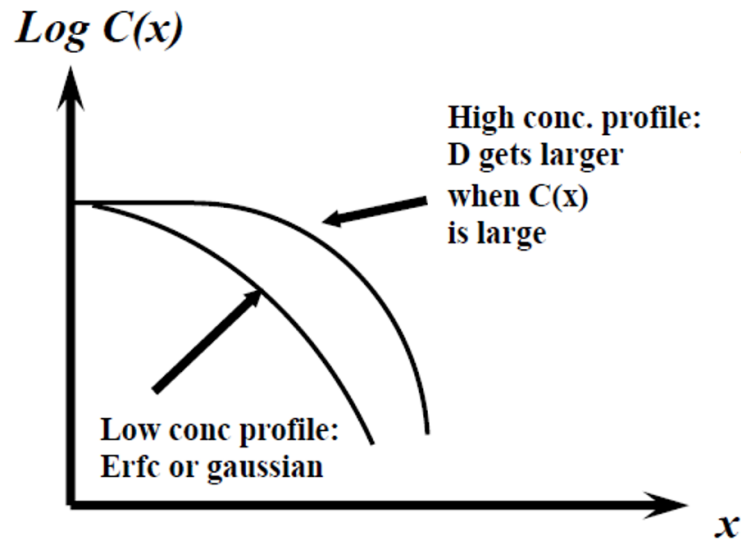
$$C(x, t) = \frac{Q}{2\sqrt{\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right)$$

$$\int C(x, t) dx = Q$$

## 4.4 Field enhanced diffusion



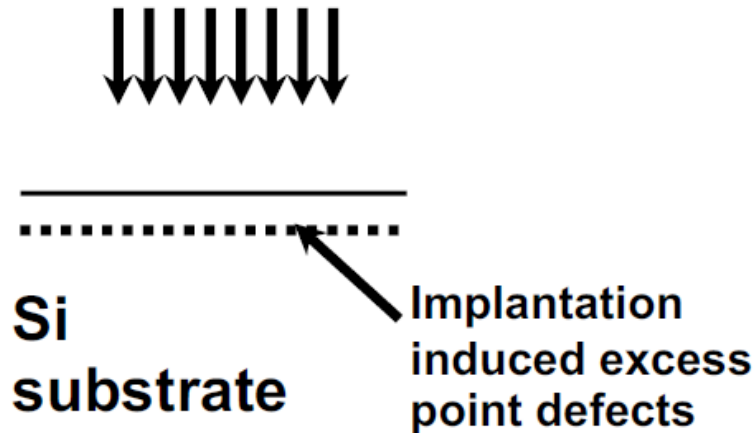
When the doping is higher than  $n_i$ , Electric field effects become important



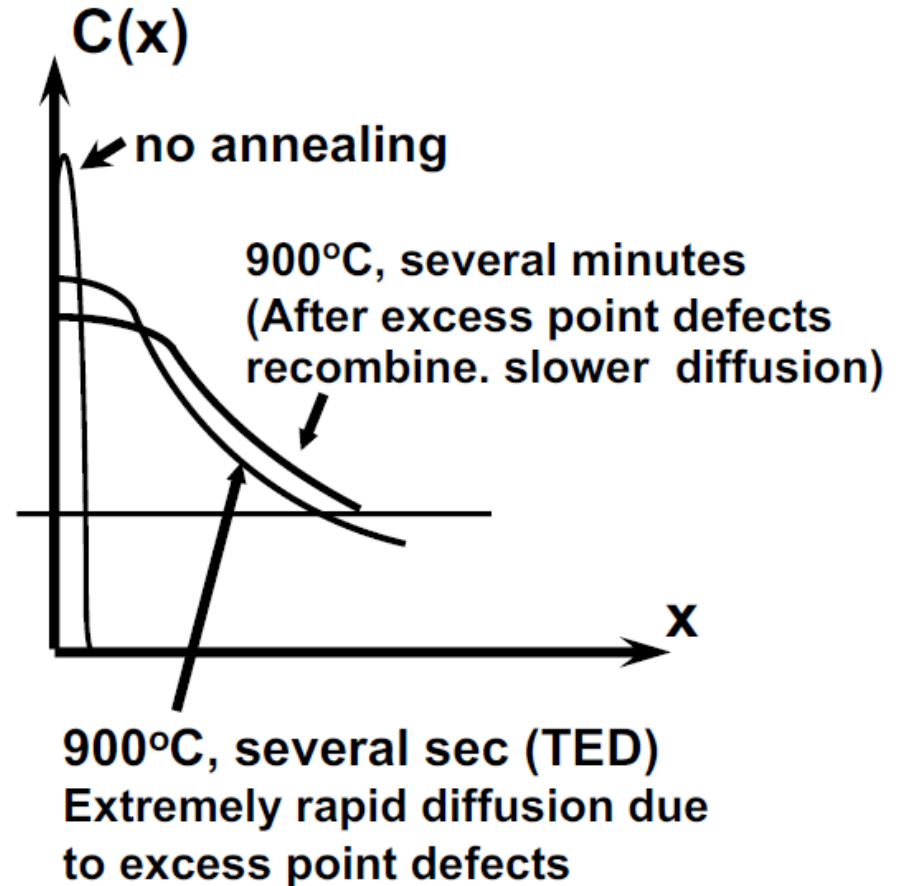
 Enhanced Diffusion for B<sup>-</sup> acceptor atoms

# 4.5 Transient Enhanced Diffusion

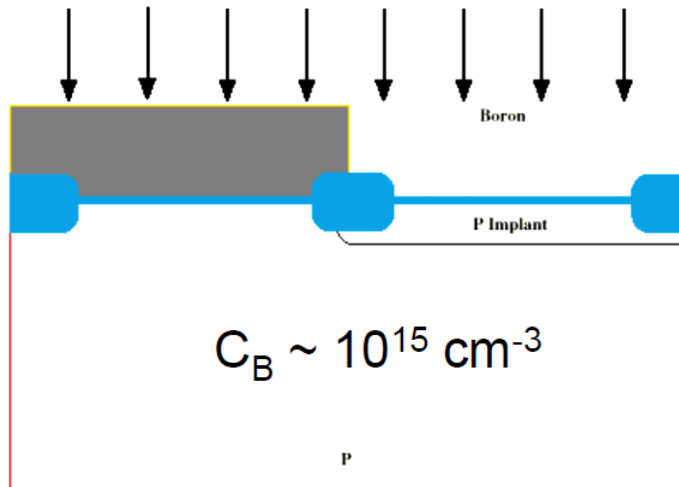
## Dopant Implantation



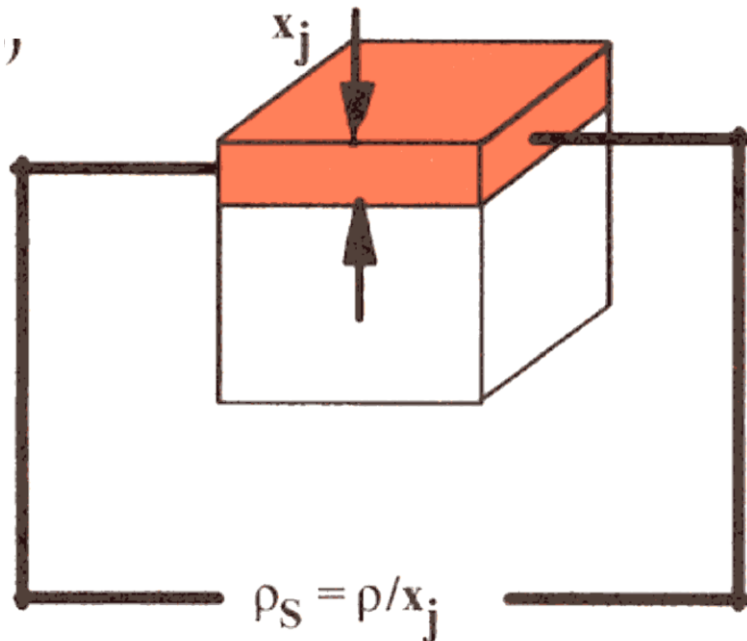
Implantation creates large number of excess Si interstitials and vacancies. (1000X than thermal process). After several seconds of annealing, the excess point defects recombine.



**Design the Drive-in Conditions (i.e. the temperature and time) for a CMOS P-well**



**Want sheet resistance  
=900 ohm/square  
And  $X_j = 3 \text{ } \mu\text{m}$**



**The sheet resistance of a shallow junction with uniform doping**

$$R = \frac{\rho}{x_j} \Omega / \text{Square} \equiv \rho_s$$

In general:  $\rho(x) = \frac{1}{q n(x) \mu[n(x)]}$   
(ohm-cm)

**For a non-uniformly doped layer**

$$\rho_s = \frac{\bar{\rho}}{x_j} = \frac{1}{q \int_0^{x_j} [n(x) - C_B] \mu[n(x)] dx}$$

This equation has been integrated numerically (by Irvin) for several analytical profiles (e.g. Gaussian and erfc profiles)



- Example of Irvin's curves below, in this case for P type Gaussian profiles.

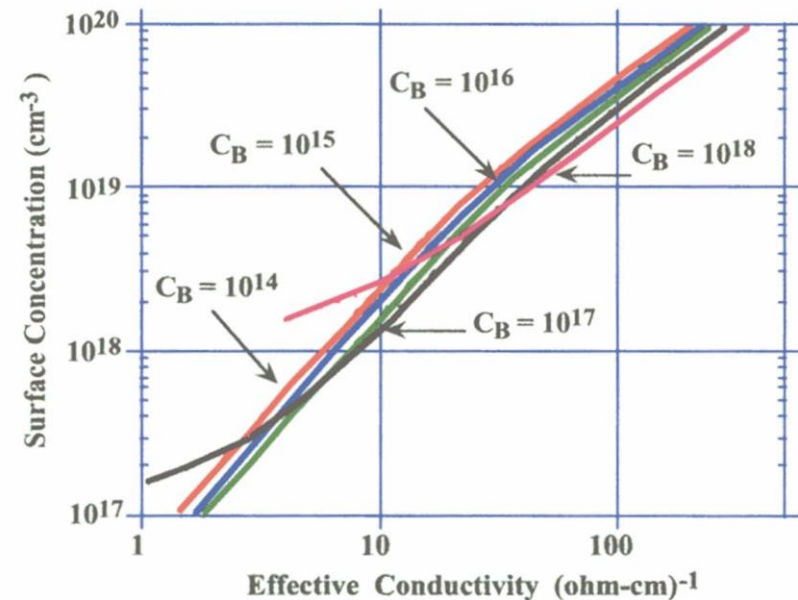
$$\rho_S = 900 \, \Omega/\text{square}$$

$$x_j = 3 \, \mu\text{m}$$

$$C_B = 1 \times 10^{15} \, \text{cm}^{-3} \text{ (substrate concentration)}$$

- The average conductivity of the layer is

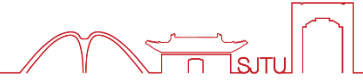
$$\bar{\sigma} = \frac{1}{\rho_S x_j} = \frac{1}{(900 \, \Omega/\text{sq})(3 \times 10^{-4} \, \text{cm})} = 3.7 \, (\Omega \cdot \text{cm})^{-1}$$



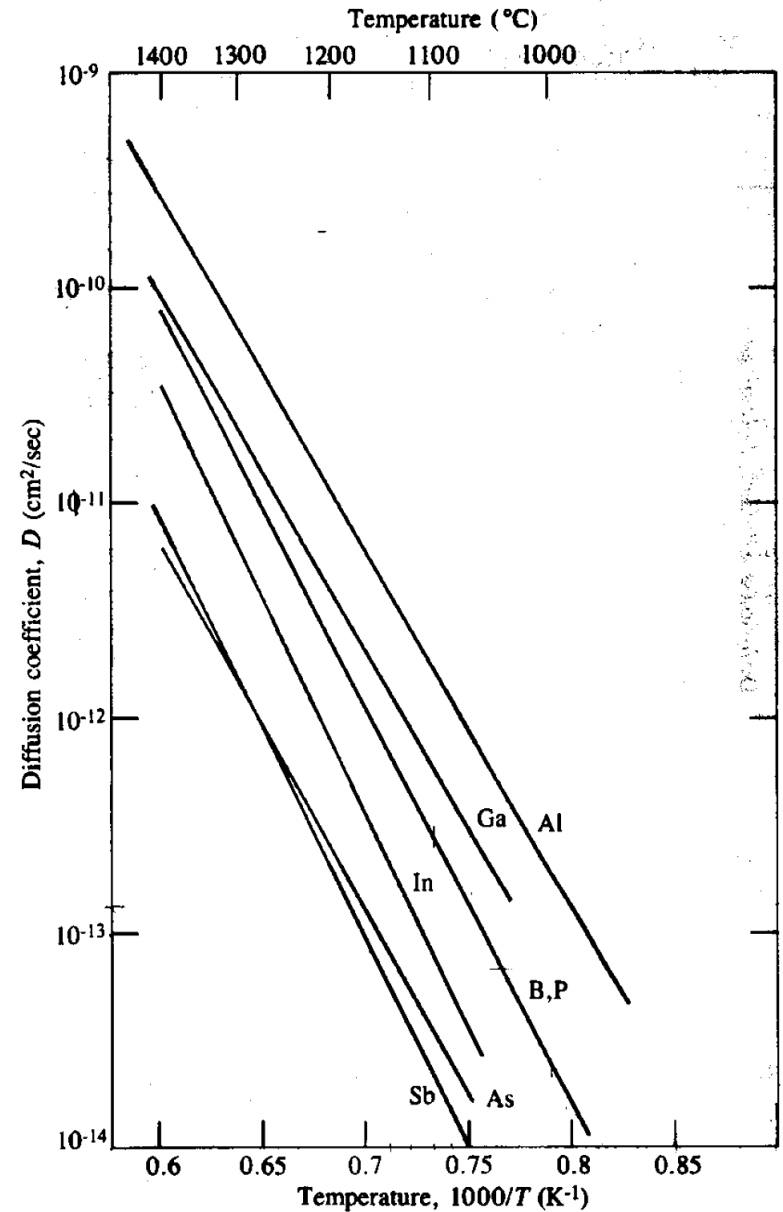
- From Irvin's curve we obtain (assuming Gaussian)

$$C_S \approx 4 \times 10^{17} / \text{cm}^3$$

## 4.6.2 Homework



$$C(x, t) = \frac{Q}{2\sqrt{\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right)$$



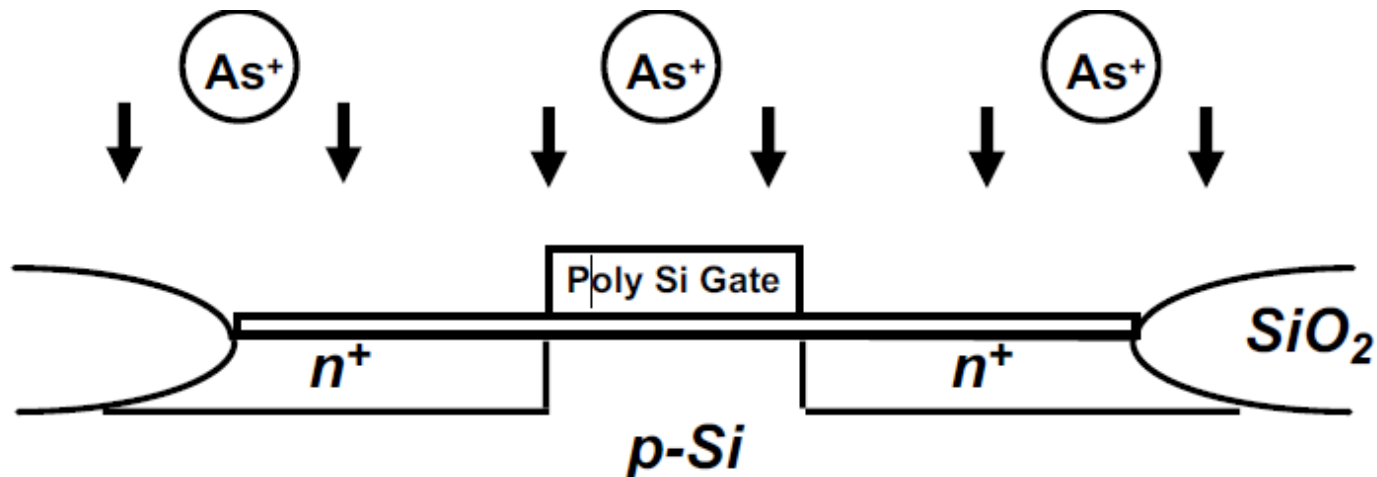
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# 5. 1 Advantages of ion implantation



- Precise control of dose and depth profile
- Low--temperature process
- Wide selection of masking materials  
e.g. photoresist, oxide, poly--Si, metal
- Excellent lateral uniformity (< 1% variation across 12" wafer)
- Less sensitive to surface cleaning procedures

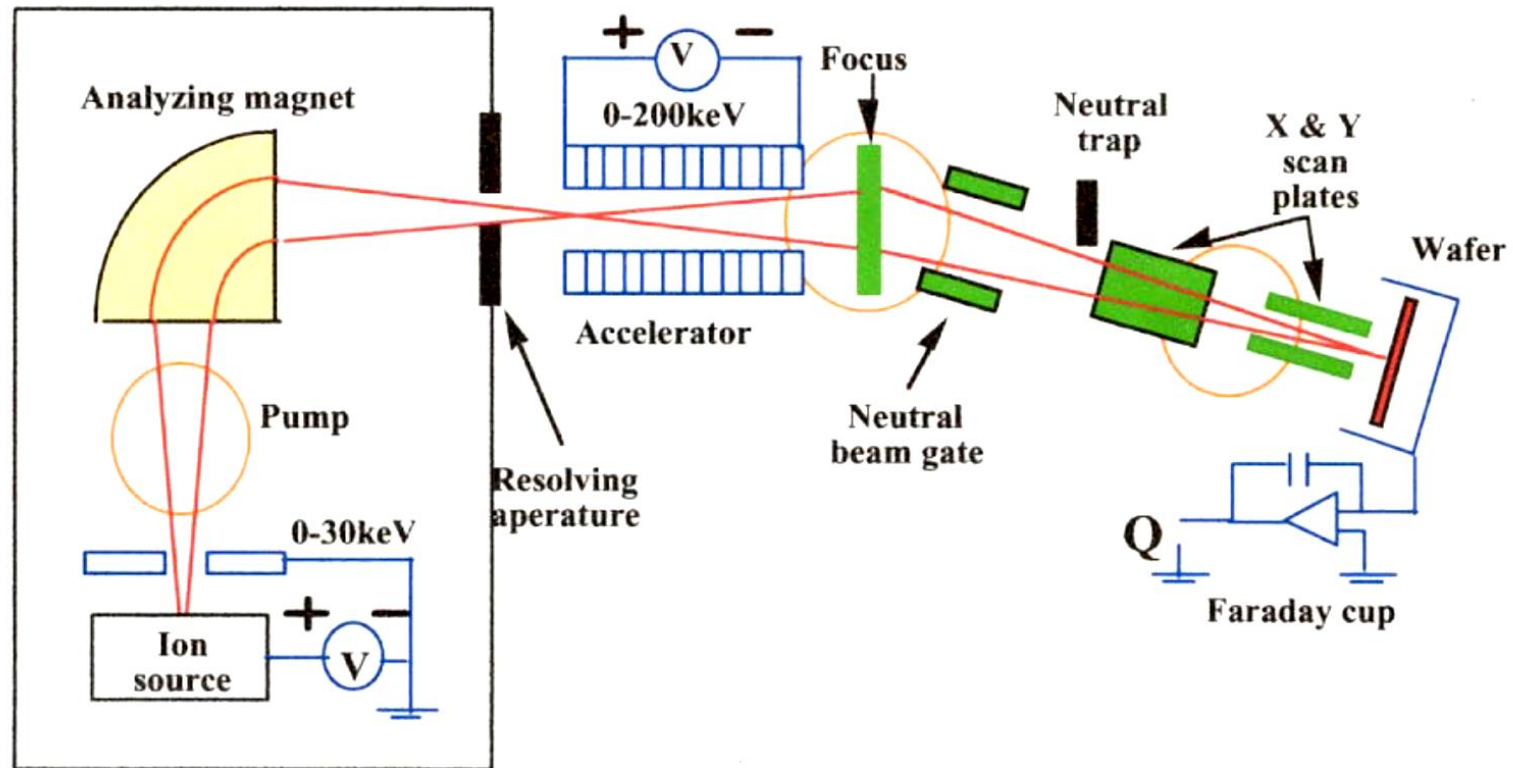
Application example: self--aligned MOSFET source/drain regions



## 5.2 Schematics of implanter



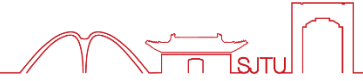
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**Implanted Dose**

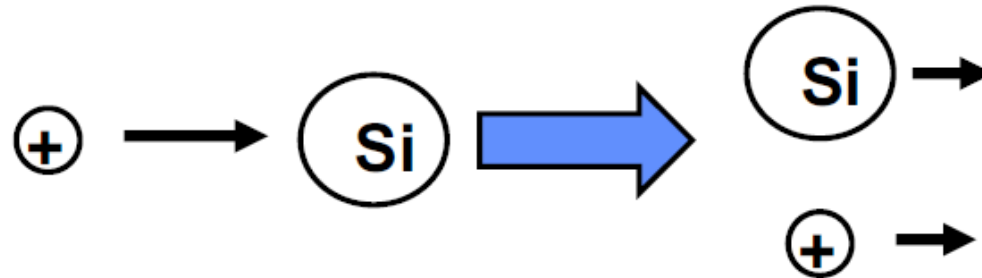
$$Q = \frac{1}{nqA} \int_0^T I(t) dt$$

## 5.3 Ion Energy Loss Mechanisms



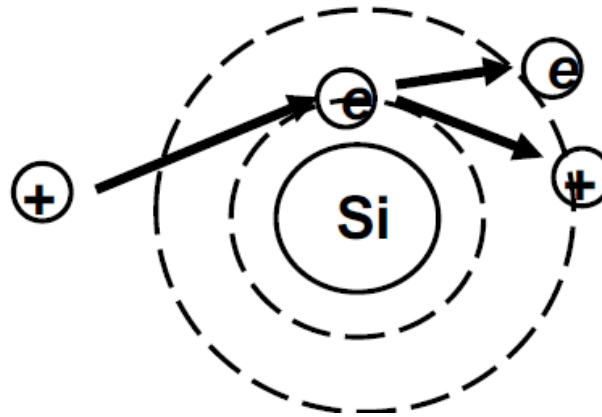
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**Nuclear  
stopping**



*Crystalline Si substrate damaged by collision*

**Electronic  
stopping**

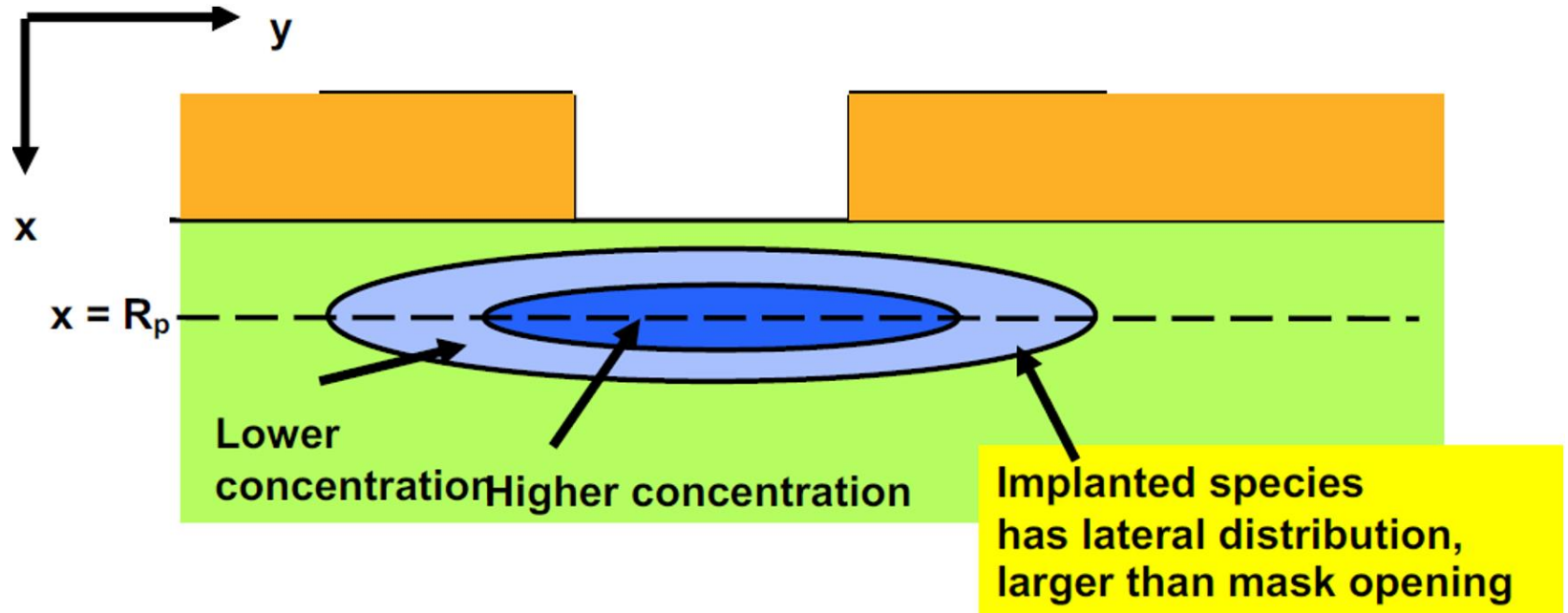
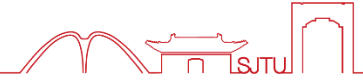


*Electronic excitation creates heat*

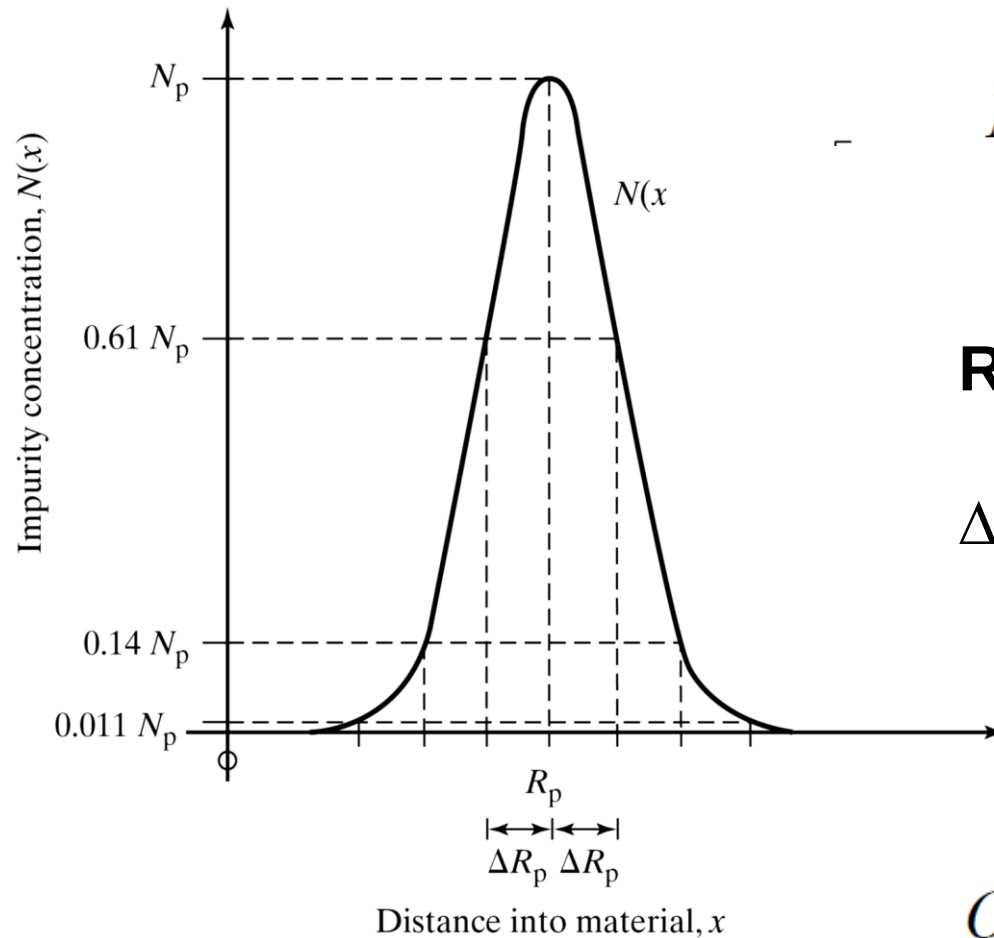
Light ions/at higher energy → more electronic stopping

Heavier ions/at lower energy → more nuclear stopping

## 5.3 Distribution of ions



# 5.3 Model for blanket implantation



$$N(x) = N_p \exp \left[ -\frac{(x - R_p)^2}{2\Delta R_p^2} \right]$$

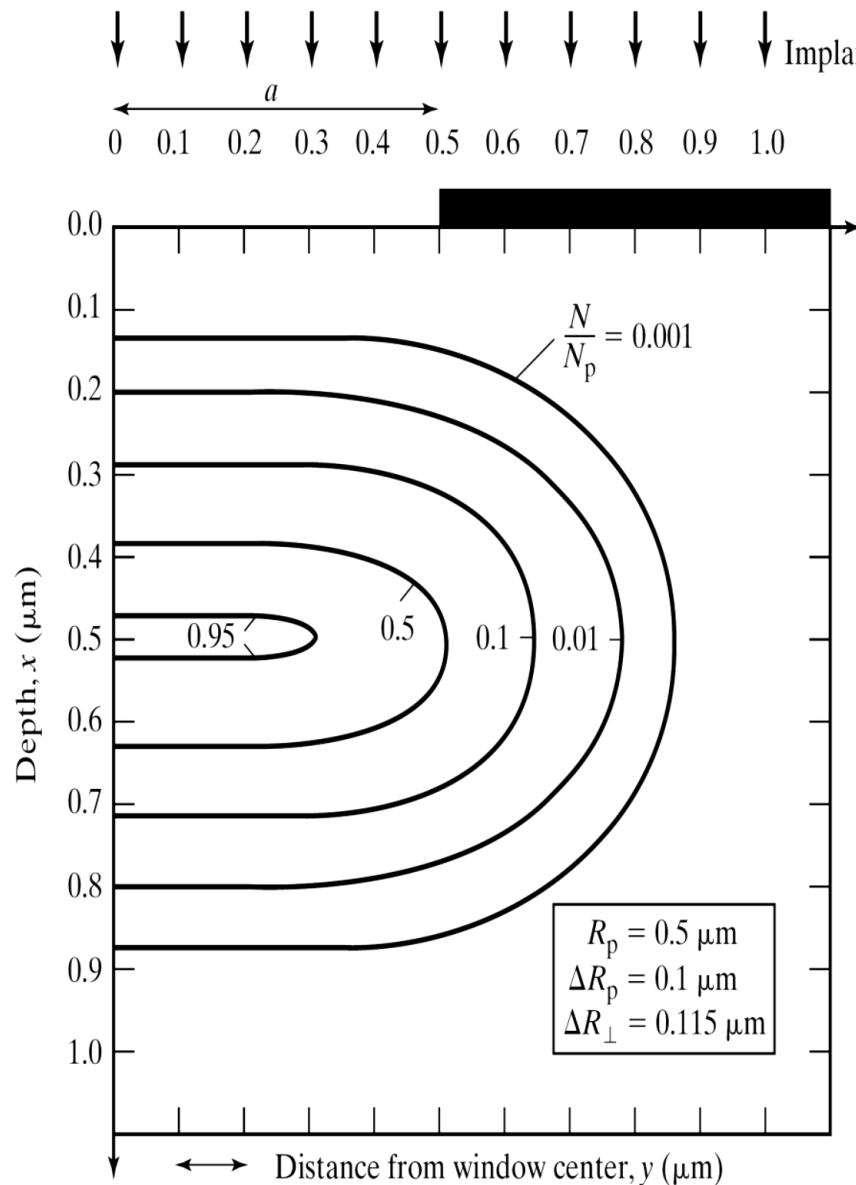
**$R_p$ : Projected Range (射程)**

**$\Delta R_p$ : Straggle (标准偏差)**

$$Q = \int_0^{\infty} N(x) dx = \sqrt{2\pi} N_p \Delta R_p$$



# 5.4 Selective ion implantation



$$N(x, y) = N(x)F(y)$$

$$F(y) = \frac{1}{2} \left[ \operatorname{erfc} \left( \frac{y-a}{\sqrt{2}\Delta R_{\perp}} \right) - \operatorname{erfc} \left( \frac{y+a}{\sqrt{2}\Delta R_{\perp}} \right) \right]$$

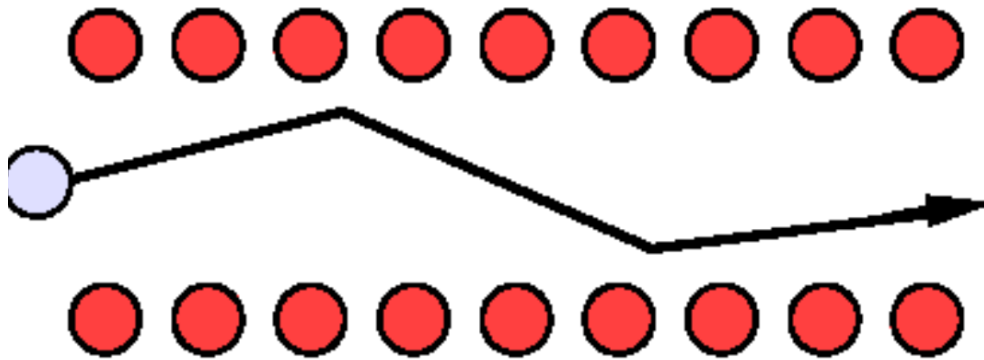
$\Delta R_{\perp}$  = transverse straggle 横向标准偏差

$N(x)$  is one-dimensional solution

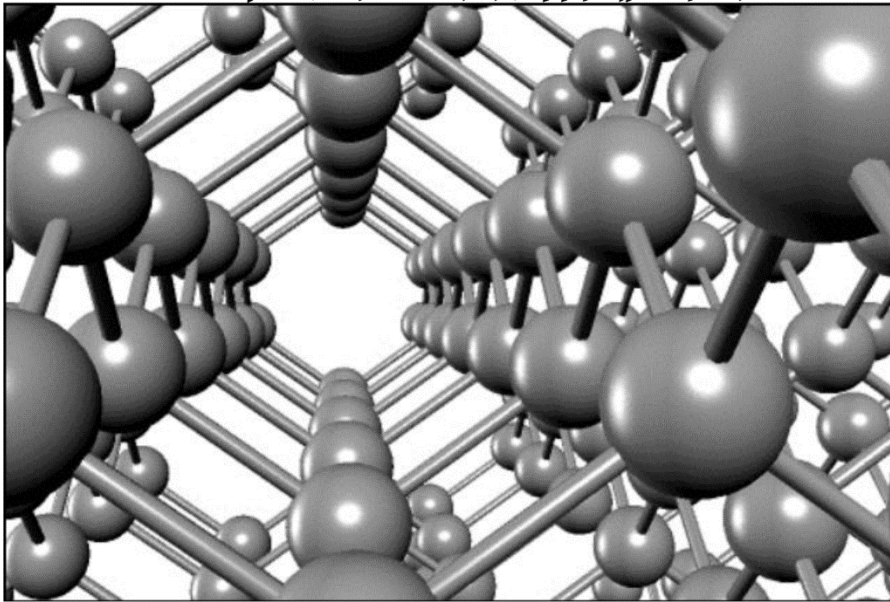
Complementary error function:

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-t^2} dt$$

# 5.5 Channeling effect

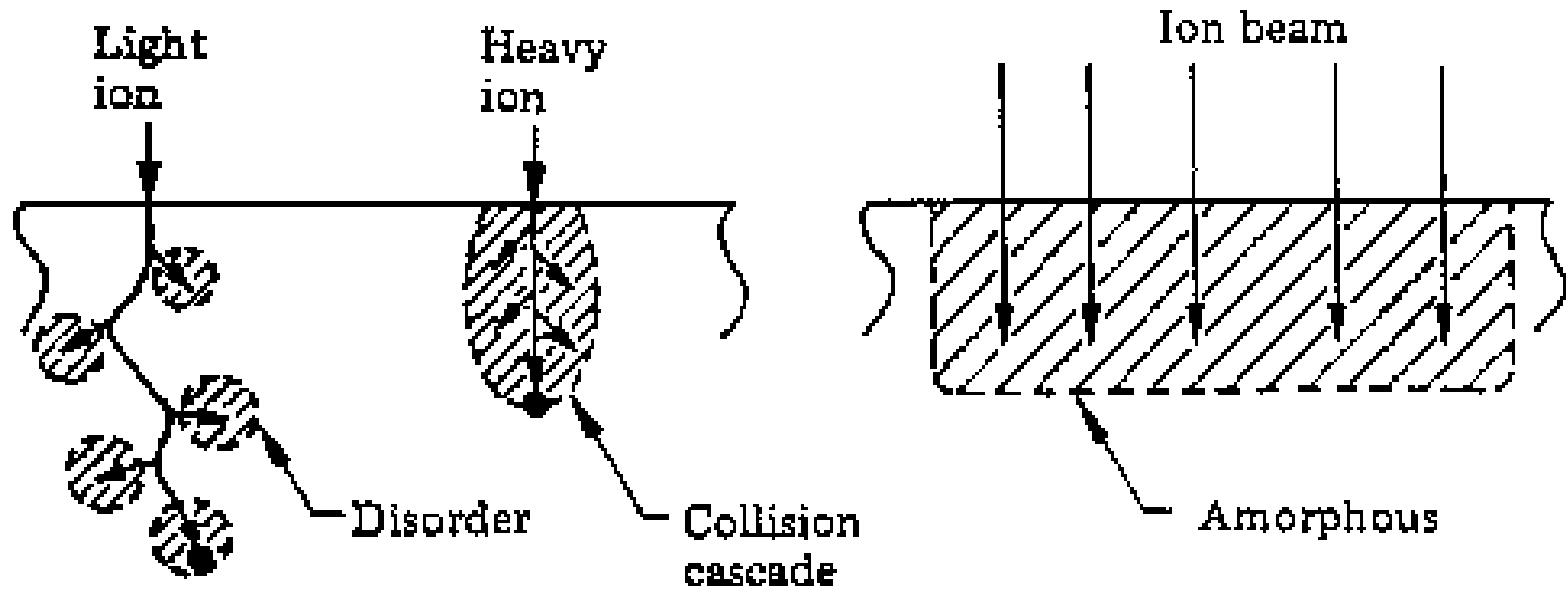


When ions are implanted along the direction of crystal axis, they cannot be scattered by the nucleus. So, the direction of the ions does not change so much that they can go ahead farther.



## Method to reduce

- Use of tilt to reduce channeling
- Amorphous layer is used
- Prevention of Channeling by Pre-amorphization

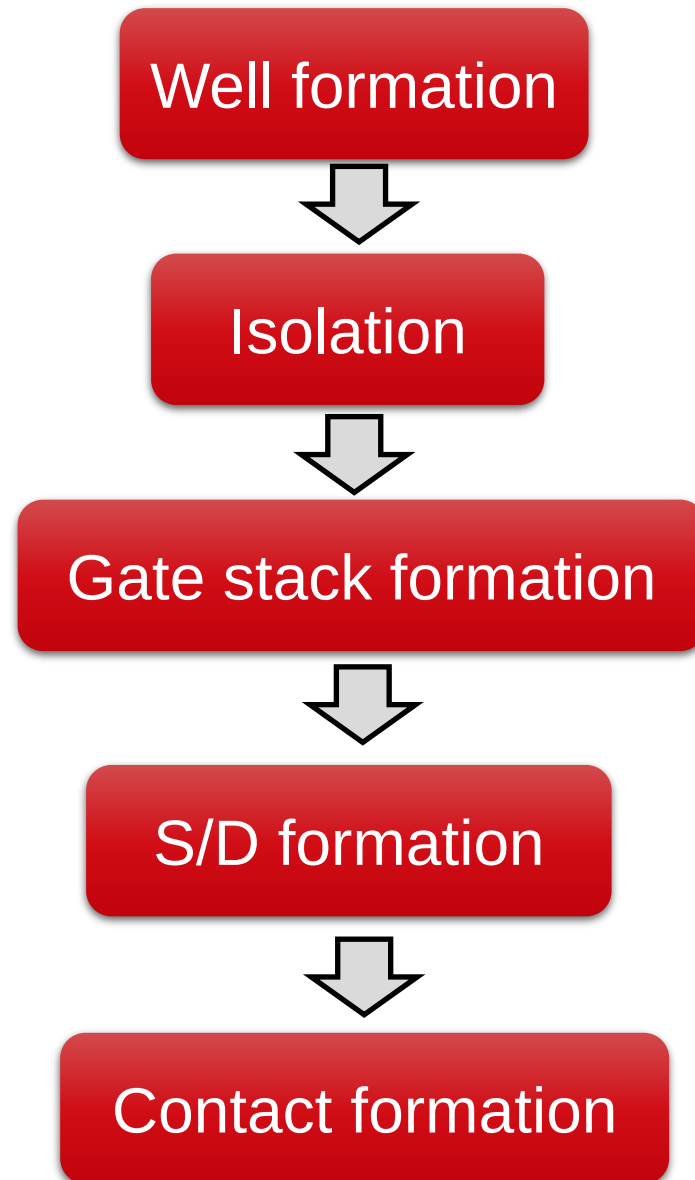


Dislocation and collision cascade are produced in Si crystal in ion Implantation causing Si amorphization with high enough doses

### Post-Implantation Annealing Summary

1. Restore Si crystallinity.
2. Place dopants into Si substitutional sites for electrical activation

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- **Basics of CMOS process**

**Substrate selection, well designing, isolation**

- **Single-well process**

**Well formation, isolation, gate stack formation, S/D formation, contact formation and connection**

- **Twin-well process**

- **Modern CMOS process**

**Method to suppress short channel effect,  
strained-Si, high-k metal gate, FinFET**

## CVD动力学描述

- Grove模型：基于反应物的运输和化学反应的模型
- CVD过程主要受两个因素控制：
  - 气相输运过程控制（反应剂在边界层中的输运）
  - 反应剂在衬底表面的化学反应（成核控制）
    - 衬底对成核的影响、化学势的影响
      - $N_1$ 表示形成单位体积薄膜所需要的原子数量
      - $h_g$ -气相质量输运系数（正比于流速的平方根）
      - $K_s$ -表面化学反应速率常数
      - $C_g$ -气体中反应剂浓度（Y代表反应物的摩尔百分比）
      - $G$ -薄膜沉积速率

$$G = \frac{F}{N_1} = \frac{k_s h_g}{k_s + h_g} \frac{C_g}{N_1}$$

## 教育和工作经历

2012.10-2015.09 [日本]东京大学 工学博士 (导师: Akira Toriumi教授)  
2015.12-2018.01 [美国]罗格斯大学 博士后研究员  
2019-至今 上海交通大学 长聘教轨副教授  
2020 入选国家级高层次海外人才计划青年项目



## 研究方向和成果

面向集成电路的新型MOS器件和功能薄膜器件, 包括高迁移率半导体器件, 高功率MOS器件, 钪基铁电器件等。迄今为止, 在Nature communication、IEDM、Electron Device Letter\Applied Physics Letter等领域内知名期刊和会议发表文章30余篇。

## 项目主持经验

前后主持国家自然科学基金“后摩尔新器件基础研究”重大研究计划培育项目、面上项目、青年项目、国家自然科学基金-日本学术振兴会合作交流项目, 华为合作课题, 上海市自然科学基金等研究课题。

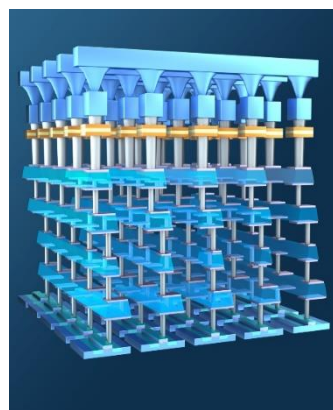
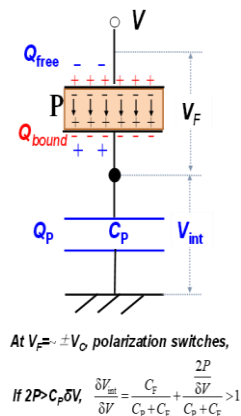


## 先进CMOS器件及集成

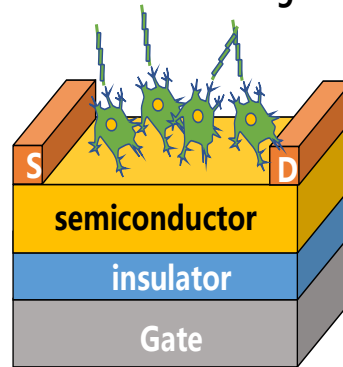
逻辑

存储

感知



Electric sensing



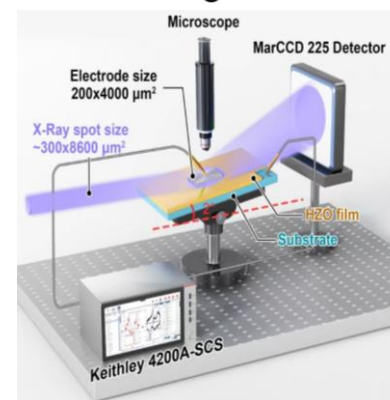
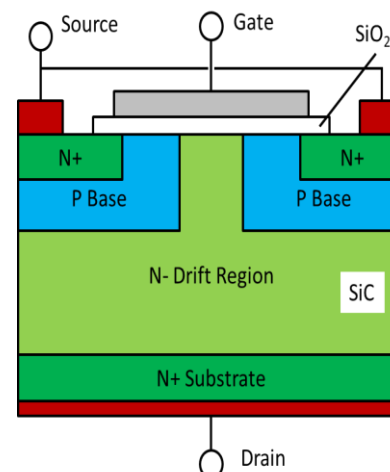
低功耗

高速度

高可靠

高灵敏

## 先进功率器件



# 谢谢

